

Article

Multi-Criteria Evaluation of Transportation Management System (TMS) Software: A Bayesian Best–Worst and TOPSIS Approach [†]

Cengiz Kerem Kütahya ¹, Bükra Doğaner Duman ^{2,*}  and Gültekin Altuntaş ³ 

¹ Aviation Academic Programs, School of Aviation, Australian University, West Mishref, Safat 13015, Kuwait; c.kutahya@au.edu.kw

² Department of Transportation and Logistics, Istanbul University, Istanbul 34000, Turkey

³ Department of Logistics, Istanbul University, Istanbul 34000, Turkey; altuntas@istanbul.edu.tr

* Correspondence: bukra.doganer@gmail.com

[†] This paper is a revised and expanded version of a paper entitled ‘A Model Proposal for Identifying and Classifying Criteria Used in the Selection of Transportation Management System (TMS) Software’, Presented at the 8th National Congress of Transportation and Logistics, Zonguldak, Turkey, 13–14 December 2024.

Abstract

Transportation Management Systems (TMSs) play a pivotal role in streamlining logistics operations, yet selecting the most suitable TMS software remains a complex, multi-criteria decision-making problem. This study introduces a hybrid evaluation framework combining the Bayesian Best–Worst Method (BBWM) and TOPSIS to identify, weigh, and rank software selection criteria tailored to the logistics business. Drawing on insights from 13 logistics experts, five main criteria—technological competence, service, functionality, cost, and software developer (vendor)—and 16 detailed sub-criteria are defined to reflect business-specific needs. The core novelty of this research lies in its systematic weighting of TMS software criteria using the BBWM, offering robust and expert-driven priority insights for decision makers. Results show that functionality (26.6%), particularly load tracking (35.8%) and cost (22.7%), mainly software license cost (39.8%), are the dominant decision factors. Beyond operational optimization, this study positions TMS software selection as a strategic entry point for sustainable digital transformation in logistics. The proposed framework empowers business to align digital infrastructure choices with sustainability goals such as emissions reduction, energy efficiency, and intelligent resource planning. Applying TOPSIS to a real-world case in Türkiye, this study ranks software alternatives, with “ABC” emerging as the most favorable solution (57.2%). This paper contributes a replicable and adaptable model for TMS software evaluation, grounded in business practice and advanced decision science.



Academic Editor: Antonio Comi

Received: 14 April 2025

Revised: 28 May 2025

Accepted: 12 August 2025

Published: 26 August 2025

Citation: Kütahya, C.K.; Doğaner Duman, B.; Altuntaş, G. Multi-Criteria Evaluation of Transportation Management System (TMS) Software: A Bayesian Best–Worst and TOPSIS Approach. *Sustainability* **2025**, *17*, 7691. <https://doi.org/10.3390/su17177691>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: Transportation Management System (TMS); Bayesian Best–Worst Method (BBWM); TOPSIS; logistics; multi-criteria decision making (MCDM); digital transformation; sustainability; green logistics

1. Introduction

The contemporary business environment is marked by complexity, uncertainty, and new forms of globalization—fueled by rapid global integration, market internationalization, advances in information and communication technologies, and rising stakeholder expectations [1] while simultaneously confronting an unpredictable competitive landscape characterized by continuous, often frame-breaking change, accelerated technological developments, faster decision-making processes, and shorter product life cycles [2].

Within this dynamic context, businesses must address multifaceted challenges regardless of their size, scale, and/or business. These include reducing operational costs, shortening production lead times, minimizing inventory levels, diversifying product offerings, expediting delivery processes, enhancing customer service quality, and improving product standards [3]. To remain competitive, businesses must effectively orchestrate global demand, supply, and production across their supply chains [4]. Also, digital transformation has emerged as a critical driver in the logistics business, not only improving operational efficiency and responsiveness but also enabling business to integrate sustainability objectives into their core processes.

To achieve these objectives, businesses must continuously transform, revising and enhancing their processes and practices while establishing an integrated model involving all actors along their supply chains [5]. Such integration demands the sharing of previously protected proprietary information with suppliers, intermediaries, distributors, and customers [6]. In addition, businesses must optimize their internal functions to generate and leverage accurate, real-time data [7]. A pivotal enabler of such an integration is the implementation of robust software systems [8].

Software systems function as the operational backbone of modern businesses. To sustain competitiveness, businesses must adopt technologies that align with strategic goals, operational priorities, and business policies, while ensuring regular updates to maintain system efficacy [9]. Compatibility with existing infrastructure and information resources is equally critical [10]. However, software selection poses significant challenges, requiring careful evaluation of multiple, often competing, elements to address businesses' needs and constraints [11].

In response to these challenges, scholars have proposed diverse frameworks to guide software evaluation and selection. The extant literature identifies numerous criteria to support decision making for both general-purpose software and domain-specific systems, such as enterprise resource planning (ERP), customer relationship management (CRM), and warehouse management system (WMS) solutions [12,13]. However, while these frameworks establish foundational evaluation criteria, they exhibit critical limitations in addressing business-specific requirements and advancing methodological sophistication. Prior research has predominantly emphasized generic or enterprise-wide systems, such as ERP [12,13], largely neglecting the specialized operational demands of niche industries.

This study addresses three key gaps in the literature. First, it reorients focus from generic ERP selection models to the transportation business, where software functionality must adapt to heterogeneous operational requirements—such as route optimization, freight management, and real-time tracking—which are inherently shaped by service variability [14]. Second, it rectifies the absence of standardized evaluation frameworks for transportation software, a methodological shortfall that impedes systematic comparisons between competing solutions and undermines strategic decision making [15]. Third, it introduces methodological innovation by integrating the Bayesian Best–Worst Method (BBWM) with a five-point Likert scale. Unlike the traditional Analytic Hierarchy Process (AHP), which necessitates exhaustive pairwise comparisons, the BBWM streamlines decision complexity while leveraging probabilistic modeling to derive criterion weights [16,17]. This hybrid approach—termed the five-point Likert scale BBWM—enhances evaluative transparency for practitioners and analytical rigor for researchers, thereby offering a replicable framework for future inquiry. In addition, the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) will be applied to the criteria weighed by the BBWM to rank two Transportation Management System (TMS) software alternatives for a real logistics business.

This study further seeks to address the following research questions:

- RQ1. Which criteria are recommended in the literature for software selection?
- RQ2. Which methodological approaches can effectively identify and prioritize business-specific criteria?
- RQ3. Which criteria should guide the selection of TMS software?
- RQ4. What are the most prevalent methods employed in software selection processes?
- RQ5. Are there alternative methodologies applicable to determining criteria for software selection?
- RQ6. What are the relative importance levels and weights of the criteria used in TMS software selection?

To address these questions, this research pursues three primary objectives: (1) to resolve existing gaps in the literature through a comprehensive review; (2) to identify and weight the criteria prioritized by transportation businesses operating in Türkiye when selecting software solutions; (3) to introduce the BBWM paired with a five-point Likert scale; and (4) to apply TOPSIS to the criteria weighed by the BBWM to rank two TMS software alternatives to be selected by a logistics business in a real case. This hybrid methodology enhances analytical rigor by simplifying complex decision-making scenarios, offering researchers a structured, probabilistic framework to derive criterion weights while maintaining interpretability for practitioners. While the literature primarily focuses on operational and cost-based considerations in software selection, this study takes a broader perspective by recognizing the strategic role of TMSs in enabling sustainable digital transformation. Logistics business are increasingly required to meet sustainability targets, including carbon emission monitoring, green route planning, and energy-efficient asset management. As digital tools such as TMSs become embedded in core logistics functions, their alignment with environmental and social responsibility frameworks becomes not only desirable but essential. Hence, evaluating TMS software through a sustainability lens—considering not only what software does, but how it contributes to long-term sustainable value creation—is a critical extension of current decision-making models.

2. Literature Review on the Selection and Decision-Making Processes of Software

The selection of any software and its associated decision-making processes have been extensively examined within the domain of multi-criteria decision making (MCDM). Software systems play a pivotal role in integrating complex operational processes across departments. However, evaluating and selecting such systems constitutes a multifaceted and inherently uncertain endeavor, necessitating careful consideration of numerous interdependent criteria as well as quantitative methods with a long process, which can be addressed from two different perspectives: (a) criteria-oriented and (b) method-oriented.

2.1. Criteria-Oriented Perspective

A criteria-driven approach addresses the challenge of selecting optimal software solutions by prioritizing the identification of business-specific evaluation metrics, rather than relying exclusively on mathematical models. Software selection represents a strategic decision with profound implications for business performance; suboptimal choices may lead to operational inefficiencies, financial losses, and diminished competitive advantage [18]. Consequently, businesses must refine their procedural frameworks for software selection by rigorously assessing both software capabilities and supplier attributes. This strategic alignment is critical for reducing operational costs, enhancing competitiveness, and fostering long-term business success.

Although both ERP and TMS software are evaluated through multi-criteria decision-making frameworks, the nature of these systems diverges significantly. ERP systems are

designed to integrate and automate enterprise-wide processes—such as finance, human resources, and inventory—requiring a focus on internal standardization, long-term scalability, and cross-departmental compatibility [19]. In contrast, TMS software is primarily operational and externally oriented, focusing on real-time logistics execution, carrier interaction, freight visibility, and routing efficiency. Furthermore, while ERP evaluations often prioritize cross-functional integration, TMS selection must account for transport mode diversity, geo-specific regulatory constraints, and dynamic route planning. These functional differences necessitate unique evaluation criteria and justify the methodological modifications employed in this study.

To operate such an approach, businesses employ a diverse array of evaluation criteria. These include usability (e.g., ease of use, interface design), technical performance (e.g., system speed, reliability, security), functional adequacy (e.g., satisfaction of needs, file conversion capabilities), and vendor-related elements (e.g., technical competence, training resources, cost-effectiveness) (Efe, 2016) [20].

Building on these foundational principles, the selection of TMS software emerges as a critical determinant of efficiency in logistics and supply chain operations. Robust TMS software can streamline route optimization, freight management, and real-time tracking, thereby reducing costs and improving customer satisfaction. However, aligning TMS software solutions with business objectives requires a structured evaluation process guided by comprehensive criteria that reflect business-specific demands. The following subsections synthesize the extant literature to delineate key criteria for TMS software selection, emphasizing their practical relevance and methodological rigor.

In recent years, the Bayesian Best–Worst Method (BBWM) has gained increasing attention as a robust approach for multi-criteria decision making under uncertainty. While the original BBWM framework was proposed by [17], recent studies have extended and applied the model across diverse domains. For instance, ref. [21] utilized the BBWM in combination with GIS for avalanche risk assessment, highlighting its adaptability in environmental hazard contexts. Similarly, ref. [22] integrated the BBWM with SERVQUAL to evaluate emergency healthcare service quality, demonstrating the method's flexibility in service settings.

In industrial applications, the BBWM has been combined with other decision-making techniques such as F-VIKOR for supplier selection in the oil and gas business [23]. Additionally, probabilistic adaptations of the BBWM for group decision making have been discussed by [17], providing further methodological refinement. The authors of [24] offer a comprehensive review of the BBWM's evolution, particularly its applications in the transportation and logistics business, and emphasize the method's ability to incorporate uncertainty, expert consensus, and hierarchical structures.

These recent contributions support the use of the BBWM in our study as both a theoretically grounded and practically validated method. Our application in the context of TMS software evaluation aligns with this growing literature and contributes to its expansion by addressing a gap in transportation software selection methodologies.

2.1.1. Functional Capabilities

Functional capabilities represent the foundational elements of any TMS software, as they determine the software's capacity to meet specific operational requirements. Critical functionalities encompass route optimization, carrier management, freight audit and payment, real-time tracking, and advanced analytics.

- **Route Optimization:** Efficient route planning is crucial for minimizing transportation costs and improving delivery timelines. Advanced route optimization algorithms can substantially reduce fuel consumption while enhancing delivery efficiency [25].

Furthermore, predictive analytics enable dynamic routing adjustments, allowing businesses to adapt to real-time variables such as traffic disruptions or adverse weather conditions [26].

- **Carrier Management:** Robust carrier management tools foster stronger partnerships with transportation providers by integrating features such as rate negotiation, performance monitoring, and capacity planning. These functionalities ensure reliable, cost-effective service delivery and operational scalability [27].
- **Freight Audit and Payment:** The automation of freight audit and payment processes reduces administrative errors and enhances financial transparency. Integrated TMS software solutions streamline invoicing and reconciliation workflows, yielding measurable cost savings [27].
- **Real-Time Tracking and Visibility:** Real-time tracking is indispensable for supply chain visibility and customer satisfaction. GPS-enabled monitoring systems allow businesses to mitigate delays proactively and communicate precise delivery updates to stakeholders [28].
- **Analytics and Reporting:** Advanced analytics generate actionable insights to inform strategic decision making. Customizable dashboards and performance reports facilitate trend analysis, operational refinement, and holistic supply chain optimization [29].

2.1.2. Scalability and Flexibility

Scalability and flexibility are critical to ensuring TMS software can adapt to evolving business requirements and sustain long-term growth.

- **Scalability:** As businesses scale their operations, TMS software must accommodate heightened transaction volumes and operational complexity. Scalable solutions are indispensable for preserving efficiency in dynamic supply chain ecosystems, particularly as market demands fluctuate [1].
- **Flexibility:** Global businesses require TMS software platforms capable of supporting multi-modal transportation networks and international logistics. Flexible systems must adapt to diverse transportation modes, regulatory frameworks, and cross-border operational challenges [30].

2.1.3. Integration and Compatibility

Seamless integration with existing infrastructure is vital for maintaining data consistency and operational cohesion.

- **System Integration:** Interoperability with ERP, WMS, and CRM systems ensures uninterrupted data flow across supply chain functions. Such integration enhances coordination and operational performance, as demonstrated in studies of supply chain digitization [31].
- **APIs and Middleware:** Open architectures, supported by APIs and middleware, enable customized integrations and interoperability between disparate systems. Modular design principles are central to achieving technological agility [32].

2.1.4. User Experience and Ease of Use

User-centric design is paramount to successful TMS software adoption and utilization.

- **Intuitive Interface:** A user-friendly interface reduces training time and improves adoption rates. For instance, perceived ease of use in driving technology has been significantly underscored in Technology Acceptance Model (TAM).
- **Training and Support:** Comprehensive training programs and responsive support mechanisms are essential for maximizing software efficacy. For example, ref. [33]’s

SERVQUAL model identifies reliability and responsiveness as core dimensions of service quality.

2.1.5. Cost and Return on Investment (ROI)

Cost–benefit analysis ensures TMS software investments align with financial and strategic objectives.

- **Total Cost of Ownership (TCO):** Businesses must evaluate both direct costs (e.g., licensing, implementation) and indirect expenses (e.g., maintenance, training). For instance, ref. [34]’s balanced scorecard framework may provide a base to advocate holistic assessment of technological value, such as that of a system.
- **ROI Analysis:** ROI calculations should incorporate tangible outcomes (e.g., cost reduction) and intangible benefits (e.g., customer satisfaction). Strategic alignment between technology and business goals optimizes long-term returns [17].

2.1.6. Vendor Reputation and Support

Vendor credibility and post-implementation support significantly influence implementation success.

- **Vendor Experience:** Providers with domain expertise in logistics and transportation are better equipped to deliver tailored, reliable solutions [35].
- **Customer Support:** Proactive technical assistance, regular updates, and issue resolution mechanisms are critical for operational continuity. These align with [33]’s emphasis on service reliability.

2.1.7. Compliance and Security

Regulatory adherence and data protection are non-negotiable in TMS software selection.

- **Regulatory Compliance:** TMS software platforms must comply with business-specific mandates, such as Electronic Logging Device (ELD) regulations and General Data Protection Regulation (GDPR), to mitigate legal and operational risks [16].
- **Data Security:** Robust encryption, access controls, and audit protocols safeguard sensitive information. The authors of [36] identify these measures as foundational to maintaining data integrity and confidentiality.

2.1.8. Method-Oriented Perspective

The selection of enterprise software, such as ERP, IT service management (ITSM), and Business Process Management Systems (BPMs), is a critical decision for businesses aiming to align technological capabilities with business objectives. Over the past two decades, researchers have proposed diverse methodologies to address the complexity of this MCDM problem. The following part of the review synthesizes key studies, methodologies, and criteria identified in the literature, highlighting trends and applications.

2.1.9. Methodological Approaches

The selection of business software, including ERP, ITSM, and BPMs, is a complex decision-making process that requires a structured evaluation of multiple criteria. Over the years, MCDM techniques have emerged as the dominant approach, with many studies incorporating fuzzy logic to address uncertainty in decision making. Early research primarily employed standalone methods such as the AHP and TOPSIS. However, as computational capabilities advance, researchers increasingly turn to hybrid models that integrate multiple methodologies to improve accuracy and adaptability.

Core MCDM Methods

Several MCDM techniques are commonly used in enterprise software selection:

- **AHP:** AHP is widely adopted for its structured pairwise comparisons, making it useful in various contexts, including BPMS selection [26] and IT system evaluation for medium-sized enterprises [37].
- **TOPSIS:** Often combined with AHP or fuzzy logic, TOPSIS ranks alternatives based on their relative proximity to an ideal solution, providing a practical and computationally efficient selection method [38].
- **Fuzzy Hybrid Methods:** Given the imprecise nature of software selection criteria, hybrid fuzzy methods have gained prominence. Examples include fuzzy AHP-TOPSIS [20], fuzzy Quality Function Deployment (QFD) [39], and fuzzy (Complex Proportional Assessment) (COPRAS) [40].
- **Innovative Integrations:** Recent studies have explored novel hybrid models, such as the integration of AHP with Teaching–Learning–Based Optimization [41] and Stepwise Weight Assessment Ratio Analysis (SWARA)-COPRAS [42], reflecting a growing trend toward sophisticated hybrid methodologies.

Advancements in Hybrid Approaches

As decision-making requirements have become more complex, researchers have increasingly adopted hybrid models to enhance the effectiveness of software selection frameworks. For instance, ref. [43] combines AHP and TOPSIS to evaluate ERP solutions in the textile business, while ref. [44] integrate fuzzy AHP to account for uncertainties in the decision-making process. The authors of [45] further extend this approach by incorporating QFD, fuzzy linear regression, and zero-one goal programming into a comprehensive selection framework.

Beyond traditional hybrid models, more recent studies demonstrate methodological innovations. Oglu’s series (2020–2022) illustrates a progression from fuzzy Sugeno to F-TOPSIS, highlighting the adaptability of MCDM techniques to dynamic business needs. Meanwhile, some researchers have explored less conventional approaches, such as the comparative analysis of ERP and office systems [46] and the Selection Approach for ERP Systems (SCAPE) selection procedure for ERP evaluation [47]. These studies showcase the diversity of methodologies in the field and the ongoing search for more effective decision-making tools.

The evolution of MCDM methods reflects broader advancements in computational decision making and an increasing recognition of subjectivity in software evaluation. Early studies, such as those by [48], primarily relied on standalone AHP, while post-2010 research has increasingly incorporated fuzzy logic to handle uncertainty. More recently, studies by [42] have explored advanced hybrid models, including iterative multi-criteria decision making (TODIM in Portuguese)–TOPSIS, demonstrating a continued emphasis on refining decision support methodologies. The widespread adoption of fuzzy techniques underscores the growing awareness that software selection involves subjective judgments and requires adaptable, context-sensitive frameworks.

Key Selection Criteria Used in MCDM Studies

Across the literature, certain selection criteria consistently emerge:

- **Functional Attributes:** Core elements such as functionality, reliability, usability, and efficiency are widely considered [39].
- **Cost Considerations:** Total cost of ownership, maintenance costs, and affordability significantly influence decision making [49,50].

- Vendor and Support: Elements such as vendor reputation, technical competence, and post-implementation services are crucial in long-term software sustainability [51,52].
- Technical Compatibility: The ability to integrate with existing systems, customization options, and scalability are key determinants of software effectiveness [53,54].

In addition, business-specific criteria emerge in specialized domains. For instance, hazardous industries incorporate “physical explosion distances” into selection models [52], while process management software in the textile business prioritizes user-friendliness and cloud adaptability [39].

2.2. Business-Specific Applications

Software selection varies across industries, reflecting distinct operational priorities. For instance, refs. [37,55] emphasize cost-effectiveness and vendor support, recognizing SMEs’ financial and resource constraints. Furthermore, ref. [56] highlight dynamic scheduling capabilities, while ref. [57] focuses on solid modeling software selection for design education, both of which focus on the manufacturing business. The author of [58] aligns IT service management (ITSM) software selection with IT infrastructure library (ITIL) V3 frameworks, stressing continual service improvement as a key objective for IT services. These business-specific applications emphasize the necessity of context-aware methodologies, as generic models may fail to address business-specific challenges effectively.

However, when it comes to MCDMs applied in software selection problems in logistics, there is little known to us. To date, no study has applied the BBWM to software selection, nor has any research focused on weighting criteria for Transportation Management System (TMS) evaluation, although the BWM, introduced by [16], has gained prominence as an efficient MCDM technique for criterion weighting, with many advantages over AHP—including reduced pairwise comparisons, higher consistency ratios, and enhanced reliability—empirically validated [59] in various contexts ranging from social sustainability [60] to circular supplier selection [41], energy efficiency [61], and logistics performance evaluation [62]. Thus, this study addresses these gaps by (1) introducing the BBWM—a probabilistic extension of the BWM, which models criteria as stochastic variables [17]—to software selection, and (2) developing the first structured framework for TMS criterion prioritization.

2.3. Criteria Set

Given the strategic importance of such a decision, a structured, multi-criteria evaluation framework has been developed by a panel of 13 experts comprising professionals, scholars, and consultants with an average age of nearly 31 and more than 12 years of industrial experience in logistics.

Development of Criteria Pool

To ensure a comprehensive and objective selection process, a detailed set of evaluation criteria has been identified based on an in-depth analysis of any logistics business’s operational needs. In addition, to validate and refine these predefined criteria, different TMS solutions—selected from leading business reports, market analyses, and software review platforms—have been examined to cross-check and enhance the evaluation framework. These criteria are further compared with key elements identified through a systematic literature review (regardless of ERP, CRM, WMS, or other software selection) to ensure alignment with best practices in TMS selection. Thus, the criteria pool is based on all the expressions used in the literature and personal views of experts rather than mathematical values such as the number or rate of repetitions.

Following the preparatory phase, five main and 34 sub-criteria were shortlisted to be used in the TMS software selection process (see Table 1). The panel of experts conducted a series of structured assessments, including software demonstrations, trial implementations, and face-to-face discussions with vendors. This rigorous evaluation process is aimed at ensuring that the selected TMS would effectively align with any logistics business's operational objectives while addressing business-specific challenges and scalability requirements.

Table 1. Shortlisted criteria for TMS software selection by a panel of experts.

Main Criteria	Sub-Criteria
Technological Competence	Information flow and transparency
	Ease of use
	Special reporting
	Interoperability
Service	Standardized reporting
	Pre-sale trial
	Technical maintenance
	Training
	After-sales support
	Reliability
	Error management
Functionality	Online support
	Load tracking
	Customization of the software
	Modules used in the software
	Installation capability
	Flexibility
	Language support
	Externalization
	Load routing
	Load planning
	Performance measurement
	Transportation cost management
	Carrier selection
Cost and Prices	Freight generation
	Dynamic routing
	License cost
	Module cost
Software Developer (Vendor)	Annual costs
	Update costs
	Experience
	Industrial knowledge
	References received
	Reputation

2.4. Elimination of Criteria

As seen in previous sections, there are many criteria to be used in the selection of TMS software based on either the extant literature or business reports, market analyses, and software review platforms. Although no restrictions have been imposed at the initial stage, pairwise comparisons of many different criteria in the pool are not possible at the individual level. Therefore, to eliminate the criteria, a model has been developed following the methodology applied by [63]. This model aims to purify and refine the selection criteria for TMS software by leveraging insights from the panel of experts. Each expert is asked to

evaluate the relevance of predefined main and sub-criteria for selecting TMS software using a four-point Likert scale including the following options—(1) not suitable for TMS software selection; (2) somewhat suitable for TMS software selection; (3) suitable for TMS software selection; and (4) highly suitable for TMS software selection—through a questionnaire form in a structured interview. Following the interview, the responses were aggregated by summing up the scores assigned by each participant for each criterion, enabling the identification of the most critical selection criteria based on their cumulative scores. In the final stage, the criteria were subsequently classified into five main dimensions per their aggregate scores: technological competence (3 criteria), service (4 criteria), cost (3 criteria), functionality (3 criteria), and software vendor (3 criteria), respectively. The finalized set of criteria is presented in Table 2, providing a structured framework for TMS software selection based on business expertise and empirical validation.

Table 2. Main and sub-criteria obtained from the analysis.

Main Criteria	Sub-Criteria
Technological Competence	Information flow and transparency
	Ease of use
	Customized reporting
Service	Training
	After-sales support
	Error management
	Reliability
Functionality	Load tracking
	Software customization
	Software modules
Cost and Prices	Software license cost
	Update cost
	Software module cost
Software Developer	Industrial knowledge
	References received
	Reputation

3. Methodology

3.1. BWM and BBWM

BWM is a subjective decision-making approach rooted in the AHP. Originally developed by [62], the BWM derives criterion weights through paired comparisons between the most and least preferred criteria or alternatives. Unlike matrix-based methods such as AHP, the BWM follows a vector-based approach, requiring significantly fewer comparisons—only $2n - 3$ comparisons compared to $n(n - 1)/2$ comparisons in AHP. This reduction in comparison burden enhances its applicability while maintaining higher consistency in results. Furthermore, a developed consistency ratio is utilized to assess the reliability of the derived weights. The authors argue that the BWM outperforms AHP due to its improved consistency in pairwise comparisons, making it a more efficient and reliable MCDM method [62].

A key advantage of the BWM is its reliance on integer-based comparisons, eliminating the need for fractional numbers, which are commonly used in AHP. This characteristic simplifies its practical implementation, increasing its usability across various decision-making contexts. Additionally, while traditional MCDM methods rely on a consistency ratio to assess the reliability of comparisons, the BWM inherently ensures a consistent

reliability level as an output, thereby reducing subjectivity-induced errors in the decision-making process.

The BBWM represents a statistical extension of the BWM, introduced by [17]. Unlike the traditional BWM, the BBWM conceptualizes criteria as random events and their corresponding weights as probabilities of realization [64]. In this probabilistic framework, data from pairwise comparisons are modeled using probability distributions, enabling a more nuanced and statistically grounded approach to weight determination.

One of the most notable advantages of the BBWM is its ability to capture the collective preferences of multiple decision makers in a group decision-making setting. Unlike conventional aggregation techniques (e.g., arithmetic or geometric averaging), the BBWM combines individual pairwise comparisons into a final probability distribution that represents the overall group consensus [64]. Furthermore, belief ranking is introduced, assigning a relationship and confidence level to each criterion pair. This ranking is then visualized through a weighted directed graph, which effectively illustrates the interdependencies among criteria.

The BBWM offers a superior alternative to other BWM extensions by integrating probability theory into the weight calculation process, thereby allowing for probabilistic control over criterion ranking. Although still a relatively recent development, empirical studies have already demonstrated its successful applications across diverse fields [17,64]. The application steps of the BBWM are systematically outlined in the literature, providing a structured methodological framework for researchers and practitioners aiming to implement this innovative decision-making tool [17].

Step 1: The main and sub-criteria of the problem are systematically identified and denoted as C_j where $j = 1, 2, 3, \dots, n$. These criteria represent the key elements influencing the evaluation and selection process. Simultaneously, the decision makers (DMs) involved in the assessment are defined as DM^k , where $k = 1, 2, 3, \dots, K$.

Step 2: Each DM determines the criteria they perceive as the most important (*Best*, C_B) and least important (*Worst*, C_W) based on their expertise and experience. These selections reflect the individual priorities of the decision makers in the evaluation process.

Following this identification, each DM assesses the best criterion in comparison to all other criteria. This process generates the best-to-others vector denoted by AB , which is mathematically represented as shown in Equation (1). This structured approach ensures a systematic and objective weighting of criteria, forming the foundation for further analysis within the Best–Worst Method (BWM) framework.

$$A_B^c = \{a_{B1}^c, a_{B2}^c, a_{B3}^c, \dots, a_{Bn}^c\} \quad (1)$$

Similarly, each DM assesses the worst criterion concerning all other criteria. This process results in the formulation of the others-to-worst vector denoted by A_W , which is mathematically defined in Equation (2). By incorporating both the best-to-others and others-to-worst comparisons, this approach ensures a balanced and structured weighting of criteria, enhancing the reliability of the Best–Worst Method (BWM) analysis.

$$A_W^c = \{a_{W1}^c, a_{W2}^c, a_{W3}^c, \dots, a_{nW}^c\} \quad (2)$$

In these vectors, a_{Bj} represents the degree of preference of the best criterion over the j th criterion, while a_{jW} denotes the degree of preference of the j th criterion over the worst criterion. The best criterion is always equally preferred to itself, meaning that $a_{BB} = 1$, and similarly, the worst criterion is always equally preferred to itself, ensuring that $a_{WW} = 1$. These conditions provide a consistent reference point for the paired comparisons within the Best–Worst Method (BWM) framework.

Step 3: Unlike the classical Best–Worst Method (BWM), where the A_B^k and A_W^k vectors obtained from different decision makers are directly aggregated, the Bayesian BWM adopts a statistical perspective to derive the A_B and A_W vectors. From this viewpoint, each criterion is regarded as a stochastic event, and the corresponding weights represent the probabilities of occurrence.

Per probability theory, the weight assigned to each criterion (w_j) must be greater than zero, and the total weight of all criteria in the decision problem must sum to one. This condition justifies the use of probabilistic modeling in decision making, ensuring that the weight distributions accurately reflect uncertainty and expert preferences.

To model both the input distributions (A_B and A_W) and the output distributions (i.e., the optimal integrated final weights), the multinomial distribution is employed for both the best and the worst criteria. The probability mass function for the worst criterion (A_W) is mathematically expressed in Equation (3), where w denotes the probability distribution and A_w is the number of occurrences of each event.

$$P(A_w | w) = \left\{ \left(\sum_{\{j=1\}}^n a_{\{jw\}} \right)! \right\} \left\{ \prod_{\{j=1\}}^n a_{\{jw\}}! \right\}; \prod_{\{j=1\}}^n w_j^{a_{\{jw\}}} \quad (3)$$

In the Bayesian Best–Worst Method (BBWM), the probability of event j is determined using the multinomial distribution, which models the likelihood of different criteria being selected as the best or worst by decision makers. For such a distribution, the probability of a given criterion is calculated as the ratio of its number of occurrences to the total number of trials.

Mathematically, this probability is formulated as shown in Equation (4), ensuring that the derived weights accurately reflect expert preferences while maintaining probabilistic consistency.

$$w_j \propto \left\{ a_{\{jW\}} \right\} \left\{ \sum_{\{i=1\}}^n a_{\{iW\}} \right\}, \forall j = 1, \dots, n \quad (4)$$

In a manner like the previous probability formulation, the weight of the worst criterion (w_W) is obtained based on the condition that $a_{WW} = 1$. This ensures that the worst criterion serves as a reference point in the weighting process.

Considering Equation (4), the relationship required to determine the optimal criterion weights in the BBWM framework can be formulated as shown in Equation (5). This step is fundamental in ensuring consistency in the weighting process, aligning the derived weights with decision makers' comparative evaluations of the best and worst criteria.

$$\{w_j\} \{w_W\} \propto a_{\{jW\}}, \forall j = 1, \dots, n \quad (5)$$

Like the modeling of A_W , the best-to-others vector A_B can also be represented using a multinomial distribution. However, a key distinction arises in the modeling process: the sequence of operations in the pairwise comparisons differs between the best and worst criteria. Specifically, the comparative structure is reversed when evaluating the best criterion against the others. This adjustment is mathematically expressed in Equation (6), where (w) is the probability distribution and refers to the element-wise division ensuring that the probabilistic framework accurately reflects the hierarchical preference relationships among criteria.

$$A_{B\text{-multinomial}}() \quad (6)$$

Using Equation (6), the expression in Equation (7) can be written similarly:

$$\{w_B\} \{w_j\} \propto a_{\{Bj\}}, \forall j = 1, \dots, n \quad (7)$$

The methodological approach undertaken in this study involves modeling the inputs of the Best–Worst Method (BWM) using a multinomial probability distribution. This proba-

bilistic framework enables the derivation of criteria weights in MCDM problems through the estimation of a probability distribution. Given that criterion weights must adhere to the constraints of non-negativity and summation to unity, the Dirichlet distribution (*Dir*) emerges as the most appropriate statistical model, as it inherently satisfies such properties.

To operationalize this, statistical inference techniques are employed to estimate two key components: the optimal integrated weight vector ($w^* = w_1^*, \dots, w_n^*$), representing the aggregated priorities across all criteria, and the individual weight vectors (w^k , $k = 1, \dots, K$) corresponding to each decision maker (DM), derived from the probabilistic model by using Equation (8). This approach ensures robustness in weight estimation by leveraging the Dirichlet distribution's capacity to model compositional data, thereby aligning theoretical rigor with the practical requirements of MCDM applications.

$$\begin{aligned} W^C &\sim \text{multinomial} \left(\frac{1}{w} \right), \quad \wedge c = 1, \dots, K \\ W^C &\sim \text{multinomial} (W^K), \quad \wedge c = 1, \dots, K \\ (w^*) &\sim \text{Dir}(y x w^*), \quad c = 1, \dots, K \\ Y &\sim \text{gamma}(0.1, 0.1) \\ W^* &\sim \text{Dir}(1) \end{aligned} \quad (8)$$

To estimate the optimal criteria weights, the Bayesian Best–Worst Method (BBWM) employs Just Another Gibbs Sampler (JAGS), a computational tool for Bayesian inference that utilizes Markov chain Monte Carlo (MCMC) simulation. This approach facilitates the derivation of the posterior probability distribution for the criteria weight vector (W), which is informed by the aggregated evaluations of decision makers.

The methodological framework is finalized through the application of credal ranking [17], a metric grounded in the Dirichlet distribution. Credal ranking quantifies the degree of superiority among criteria and evaluates the reliability of these dominance relationships. The results are complemented by visual representations—such as credibility intervals or posterior probability plots—to illustrate the confidence levels associated with the hierarchical ordering of criteria.

Unlike traditional MCDM methods that require a separate consistency ratio, the Bayesian Best–Worst Method (BBWM) incorporates consistency evaluation through its probabilistic structure. Specifically, the credal ranking technique, based on the Dirichlet posterior distributions, allows for the assessment of the confidence level in the dominance of one criterion over another. The resulting credibility intervals, as visualized in Figures 1 and 2, serve as statistical evidence supporting the robustness of the derived weights. This built-in mechanism offers a more nuanced and reliable approach to consistency validation in group decision-making settings, as also supported by [17].

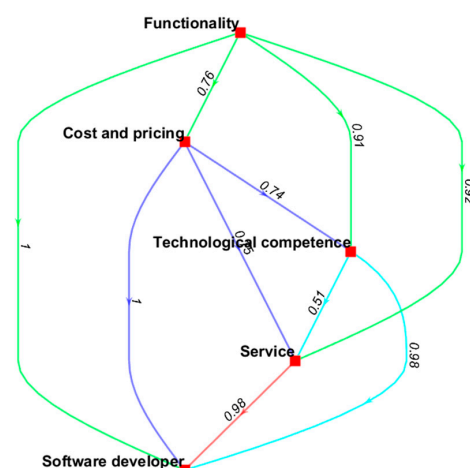


Figure 1. Credal ranking display of main criteria.

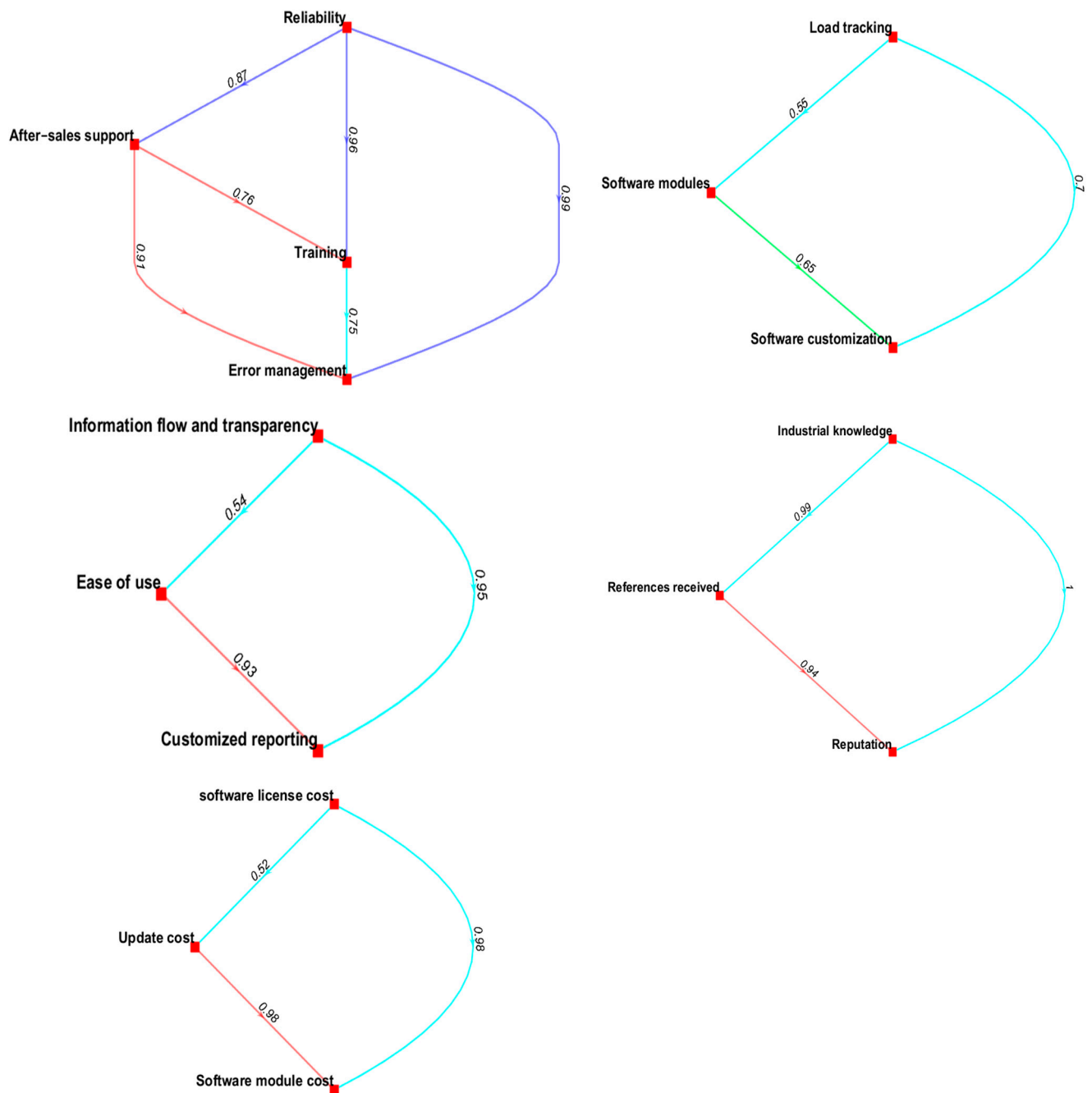


Figure 2. Credal ranking display of sub-criteria.

The Dirichlet distribution is selected as the prior for the probabilistic modeling of criteria weights in the BBWM due to its mathematical properties. Specifically, it is the natural conjugate prior for the multinomial distribution and ensures that the generated weight vectors are non-negative and collectively sum to one. Additionally, the Dirichlet distribution provides analytical convenience and computational efficiency when deriving posterior estimates, which makes it particularly suitable for group decision-making scenarios involving normalized weights.

3.2. Implementation

This section builds upon the BBWM methodology outlined earlier to evaluate the criteria for TMS software selection within an integrated decision-making framework. The hierarchical structure consists of five main criteria and sixteen sub-criteria (Table 2). The

BBWM is employed to derive the relative weights of these criteria using a probabilistic group decision-making model that incorporates expert consensus and accounts for uncertainty in preferences. The analysis proceeds in two stages: (1) pairwise evaluation of main criteria relative to one another, and (2) pairwise evaluation of sub-criteria within each main criterion. Once the weight structure is established, the TOPSIS method is applied to assess and rank the TMS software alternatives based on their performance across these weighted criteria. This sequential integration ensures a coherent transition from criteria prioritization to alternative evaluation, allowing the final decision to reflect both subjective expert judgment and objective performance metrics.

3.2.1. Data Collection and Expert Input

To determine the weights of predefined criteria for TMS software selection, a panel of eight business experts, actively engaged in logistics operations, is convened. Participants, averaging 31 years of age with six years of business-specific expertise, contribute evaluations through a structured three-phase protocol: (a) Likert scale assessments where criteria are rated on a five-point scale (1: very low to 5: very high) to quantify their perceived importance; (b) best–worst pairwise comparisons through which each decision maker (DM) identified the most (best) and least (worst) critical main criterion, followed by pairwise comparisons of the best criterion against all others; and (c) construction of two evaluation matrices (best-to-others and others-to-worst).

To ensure methodological rigor, a user-friendly decision support system is designed to aggregate responses, validate consistency, and minimize evaluator bias. The system automates data harmonization across the eight experts, consolidating inputs for both main and sub-criteria into a unified analytical framework.

3.2.2. Bayesian Computation and Results

The aggregated evaluations have been processed using matrix laboratory (MATLAB R2024b), following the Bayesian inference procedure detailed by [17]. Customized versions of the authors' publicly available MATLAB R2024b codes (URL-1, 2021) are executed to compute local weights for criteria and sub-criteria, derived from posterior distributions and belief rankings, quantifying the probabilistic dominance relationships between criteria.

The results include Table 3, presenting the BBWM-derived weights of the main criteria, and Figure 1, illustrating belief rankings via credibility intervals. These outputs validate the robustness of hierarchical prioritization, aligning theoretical rigor with empirical expert judgments.

Table 3. Main criterion weights obtained with BBWM.

No.	Main Criteria	Weight	Ranking
1	Technological Competence	0.1952	3
2	Service	0.1939	4
3	Functionality	0.2660	1
4	Cost	0.2270	2
5	Software Developer (Vendor)	0.1179	5

The analysis reveals functionality as the paramount main criterion in TMS software selection, whereas software vendor business emerges as the least influential. Notably, cost—ranked second—exhibits minimal disparity in weight scores relative to functionality, yet both criteria demonstrate substantial divergence from software vendor business. This hierarchy underscores the prioritization of software functionality over ancillary elements in business decision making.

As illustrated in Figure 1, the credal ranking derived from the Bayesian BWM delineates the probabilistic dominance relationships among criteria. Functionality occupies the apex, with directed edges (arrows) originating from it, signifying its superiority over subordinate criteria. Conversely, criteria such as software vendor business, positioned at the nadir, receive edges from all other criteria, reflecting their diminished relative importance. The width and directionality of these edges quantify the confidence intervals associated with each dominance relationship, as modeled by the Dirichlet distribution.

Table 4 delineates the weight scores of sub-criteria underpinning the TMS software selection process, derived via the Bayesian BWM. The hierarchical prioritization regarding the technological competence domain reveals that information flow and transparency attain the highest weight, emphasizing the imperative for software to enable seamless data transfer and operational clarity across stakeholders. Furthermore, ranked second, ease of use underscores the necessity of intuitive user interfaces to minimize learning curves. In addition, demonstrating comparatively lower priority, customized reporting is expected to have a secondary emphasis on tailored reporting features.

Table 4. Sub-criterion weights obtained with BBWM.

No.	Main Criteria	No.	Sub-Criteria	Weight	Ranking
1	Technological Competence	1.1	Information flow and transparency	0.3894	1
		1.2	Ease of use	0.3769	2
		1.3	Customized reporting	0.2337	3
2	Service	2.1	Training	0.2116	3
		2.2	After-sales support	0.2589	2
		2.3	Error management	0.1734	4
		2.4	Reliability	0.3560	1
3	Functionality	3.1	Load tracking	0.3584	1
		3.2	Software customization	0.2990	3
		3.3	Software modules	0.3426	2
4	Cost	4.1	Software license cost	0.3977	1
		4.2	Update cost	0.3922	2
		4.3	Software module cost	0.2101	3
5	Software Developer (Vendor)	5.1	Industrial knowledge	0.5217	1
		5.2	References received	0.2901	2
		5.3	Reputation	0.1883	3

With respect to the service domain, reliability emerges as the foremost element, reflecting the criticality of consistent software performance and uninterrupted functionality, followed closely by after-sales support, highlighting the value of sustained vendor assistance post-implementation. Additionally, assigned reduced weights, training and error management are expected to play an ancillary role relative to core service attributes.

Concerning the functionality domain, load tracking dominates as the most critical sub-criterion, prioritizing the software's capacity to monitor logistics operations in real time. Ranking second, software modules emphasize the need for modular flexibility to accommodate diverse operational workflows. Moreover, software customization displays moderate importance, aligning with businesses' demands for adaptability.

Speaking of the cost domain, software licenses claim the highest weight, underscoring the dominance of initial investment considerations, followed with notable significance by update cost, signaling concerns over long-term maintenance expenditures. In addition, software module cost is assigned to the lowest priority, suggesting less emphasis on modular pricing structures.

As for the software developer (vendor) domain, industrial know-how achieves paramount importance, stressing the necessity of vendor expertise in business-specific challenges, while references and reputation have lower weights, implying limited influence of vendor accolades relative to technical proficiency.

Figure 2 complements these findings through credal ranking displays, illustrating probabilistic dominance relationships among sub-criteria. The visualizations corroborate the hierarchy in Table 4, with arrows quantifying confidence intervals for superiority assertions. For instance, information flow and transparency exhibit robust dominance over peers in technological competence, while industrial know-how maintains clear precedence in software vendor evaluations.

3.3. TOPSIS

Leveraging the dataset derived in the previous section, a comparative evaluation was conducted between two software vendors specializing in logistics-business solutions. The analysis employs TOPSIS, a MCDM method introduced by [65]. TOPSIS operationalizes the principle of identifying alternatives that simultaneously minimize geometric distance to the positive ideal solution (PIS) and maximize distance to the negative ideal solution (NIS) [66,67].

Step 1: Objectives are set, and evaluation criteria are defined.

Step 2: A decision matrix (D) is constructed with the help of Equation (9). Alternatives ($a_1, \dots, a_n$) are systematically listed, and for each alternative, the corresponding characteristics for each criterion ($y_{1k}, \dots, y_{nk}$) are recorded, as detailed by [68].

$$D = [y_{11} \quad y_{12} \quad y_{1k} \quad \dots] \quad (9)$$

Step 3: The normalized decision matrix (R) is then constructed. This matrix is derived by normalizing the original decision matrix through computing the square root of the sum of the squares of the criterion scores or attributes [68]. Equation (10) delineates the normalization process, which ultimately yields the R matrix presented in Equation (11).

$$r_{ij} = \frac{y_{ij}}{\sqrt{\sum_{i=1}^n y_{ij}^2}} \quad i = 1, 2, \dots, n \quad j = 1, 2, \dots, k \quad (10)$$

$$R = [r_{11} \quad r_{12} \quad r_{1k} \quad \dots] \quad (11)$$

Step 4: The weighted normalized decision matrix (V) is created through Equation (12). Each criterion j is assigned a weight (w_j) reflecting its relative importance. The weights are applied to the corresponding elements of the normalized decision matrix, as delineated by [66].

$$W = [w_{11} \quad w_{12} \quad w_{1k} \quad \dots] \quad (12)$$

Subsequently, each element within the columns of the R matrix (refer to Equation (11)) is multiplied by its corresponding weight (w_{ij}) as specified in Equation (12). This operation results in the formation of the V matrix, as presented in Equation (13) [66].

$$V = [v_{11} \quad v_{12} \quad v_{1k} \quad \dots] \quad (13)$$

Step 5: Ideal (A^+) and negative ideal (A^-) solutions are obtained. The positive ideal solution is defined as the set of optimal performance values obtained from the weighted normalized decision matrix, whereas the negative ideal solution comprises the least favorable values [67]. These ideal solutions are computed using Equations (14) and (15).

$$A^+ = \{(\max_{ij} v_{ij} \mid j \in I), (\min_{ij} v_{ij} \mid j \in J)\} \quad (14)$$

$$A^- = \{(\min v_{ij} \mid j \in I), (\max v_{ij} \mid j \in J)\} \quad (15)$$

In both formulas, the symbol I represents the benefit criterion (maximization), and J represents the cost criterion (minimization) [66]. The results derived from Equation (14) can be expressed as $A^* = \{v_1^*, v_2^*, \dots, v_k^*\}$, whereas the outcomes from Equation (15) are represented as $A^- = \{v_1^-, v_2^-, \dots, v_k^-\}$.

Step 6: Discrimination measures are calculated. The separation (distance) between alternatives is measured. The distance of each alternative from the positive ideal solution is calculated as in Equation (16) [66]:

$$S_i^{\{*\}} = \sqrt{\left\{ \sum_{j=1}^n (v_{ij} - v_{ij}^{\{*\}})^2 \right\}} \quad (16)$$

Similarly, the distances from the negative ideal solution are calculated as in Equation (17) [66]:

$$S_i^{\{-\}} = \sqrt{\left\{ \sum_{j=1}^n (v_{ij} - v_{ij}^{\{-\}})^2 \right\}} \quad (17)$$

Step 7: The relative proximity to the ideal solution is calculated. The relative proximity, denoted as C_i^* , is computed using Equation (18) [66].

$$C_i^* = \frac{S_i'}{S_i' + S_i^*} \quad 0 \leq C_i^* \leq 1 \quad (18)$$

Step 8: Alternatives are ranked in accordance with their closeness C_i^* to the ideal solution. The maximum C_i^* value is selected [66].

3.3.1. TOPSIS Implementation

This section operationalizes the TOPSIS methodology, as delineated in prior sections, to evaluate and rank two leading Transportation Management System (TMS) software solutions within the logistics business in Türkiye. The analysis leverages the hierarchical criteria framework established before, comprising 5 main criteria and 16 sub-criteria (see Table 2), with their respective weights derived through the Bayesian Best–Worst Method (BBWM). The methodological integration of the BBWM and TOPSIS ensures a robust evaluation process: The BBWM-derived weights (see Table 4) reflect the probabilistic dominance relationships among criteria, grounded in expert judgments, and utilizing these weights, TOPSIS calculates the geometric proximity of each software alternative to the positive ideal solution (PIS) and negative ideal solution (NIS), as defined by [65]. This dual-phase approach harmonizes the strengths of probabilistic weighting (BBWM) and distance-based ranking (TOPSIS), enabling a systematic, transparent comparison of software alternatives against multidimensional criteria. Subsequent subsections detail the computational steps and the results of the final rankings.

Data Collection and Expert Input

The TMS software alternatives evaluated in this study are designed to support essential transportation and logistics functions across a range of operational contexts. These include load tracking, dynamic route planning, freight cost control, carrier selection, and performance analytics. The selected alternatives are commercially available and widely adopted in the logistics business. While they may vary in interface or vendor-specific features, both systems offer a comparable set of modules that address the same core operational needs. This functional alignment ensures their substitutability and justifies their comparative evaluation within the BBWM-TOPSIS framework. The selection was also

validated by expert opinion to reflect realistic and representative software options used in practice.

To evaluate TMS software alternatives using the TOPSIS, a panel of eight business experts—actively engaged in logistics operations and possessing decision-making authority—was enlisted. Participants, averaging 31 years of age with six years of business-specific experience, provided structured evaluations through a three-phase protocol: (a) a five-point Likert scale (1: very bad to 5: very good) where experts rate each software vendor’s performance across predefined main and sub-criteria; (b) criteria weighting through which experts assign relative weights to criteria using direct rating or pairwise comparisons; and (c) decision matrix construction in which individual evaluations are aggregated into rows representing alternatives (ABC and XYZ software) and columns denoting criterion-specific performance scores. In the final stage of the evaluation, two leading TMS software solutions were compared based on the developed framework. For confidentiality purposes, these systems are anonymized as ABC and XYZ. However, both are commercial off-the-shelf (COTS) TMS platforms used by medium-to-large logistics business in Türkiye. They were selected by the expert panel due to their functional alignment and prevalence in the market.

In the application of the TOPSIS methodology, expert judgment was systematically collected and processed through a structured protocol. Each of the eight participating experts independently evaluated the performance of two TMS software alternatives (ABC and XYZ) across 16 sub-criteria using a five-point Likert scale (1 = very poor, 5 = very good). The individual scores were then aggregated by calculating the arithmetic mean for each criterion–alternative pair. This aggregation step generated a consensus-based decision matrix, which was subsequently normalized and used in conjunction with BBWM-derived weights to form the weighted decision matrix. This structured approach ensures methodological consistency and enhances the reliability of the ranking results.

TOPSIS Computation and Results

The criterion weights obtained through the TOPSIS method are systematically presented in Tables 5 and 6, highlighting the hierarchical prioritization of both main and sub-criteria. While the two software vendors (assumed to be ABC and XYZ from now on) exhibit varying levels of performance across different indicators, each possessing distinct competitive advantages in specific domains, a comprehensive evaluation necessitated the application of TOPSIS.

Table 5. Decision matrix table for ABC and XYZ.

No.	Main Criteria	No.	Sub-Criteria	ABC	XYZ
1	Technological Competence	1.1	Information flow and transparency	8.12362	5.912727
		1.2	Ease of use	9.852143	6.574269
		1.3	Customized reporting	9.195982	6.295835
2	Service	2.1	Training	8.795975	5.873687
		2.2	After-sales support	7.468056	6.835559
		2.3	Error management	7.467984	5.418482
		2.4	Reliability	7.174251	5.876434
3	Functionality	3.1	Load tracking	9.598528	6.099086
		3.2	Software customization	8.803345	6.36821
		3.3	Software modules	9.124218	7.355528
4	Cost	4.1	Software license cost	1.061753	3.798695
		4.2	Update cost	3.90973	5.056938
		4.3	Software module cost	3.497328	5.369658

Table 5. *Cont.*

No.	Main Criteria	No.	Sub-Criteria	ABC	XYZ
5	Software Developer (Vendor)	5.1	Industrial knowledge	7.637017	5.139351
		5.2	References received	7.545475	6.822635
		5.3	Reputation	7.550214	5.511572

Table 6. Normalized decision matrix table for ABC and XYZ.

No.	Main Criteria	No.	Sub-Criteria	ABC	XYZ
1	Technological Competence	1.1	Information flow and transparency	0.808516	0.588474
		1.2	Ease of use	0.83181	0.555061
		1.3	Customized reporting	0.825146	0.564919
2	Service	2.1	Training	0.831627	0.555335
		2.2	After-sales support	0.737654	0.675179
		2.3	Error management	0.809394	0.587265
		2.4	Reliability	0.773609	0.633664
3	Functionality	3.1	Load tracking	0.844022	0.536308
		3.2	Software customization	0.810232	0.58611
		3.3	Software modules	0.778526	0.627612
4	Cost	4.1	Software license cost	0.963088	0.269188
		4.2	Update cost	0.791126	0.611653
		4.3	Software module cost	0.837941	0.545762
5	Software Developer (Vendor)	5.1	Industrial knowledge	0.829635	0.558305
		5.2	References received	0.741742	0.670685
		5.3	Reputation	0.807691	0.589606

As discussed in previous sections, TOPSIS is chosen for its ability to integrate multiple criteria, quantify geometric proximity to ideal solutions, and produce a unified ranking that accounts for both benefit maximization and cost minimization principles (Hwang & Yoon, 1981) [65]. This methodological approach ensures that trade-offs between conflicting criteria—such as functionality versus cost or vendor reliability versus technical adaptability—are systematically analyzed. Consequently, it facilitates a balanced, data-driven assessment of vendor suitability within the logistics business.

Table 5 presents the initial evaluation scores of the two software alternatives (ABC and XYZ) based on 16 sub-criteria, as assessed by expert opinions. The scores are assigned on a scale from 1 to 10, where higher values denote superior performance in benefit-type criteria, while lower values indicate better performance in cost-type criteria. This table serves as the foundational input for the TOPSIS analysis, providing the raw data necessary for normalization and subsequent multi-criteria evaluation.

The normalized decision table (Table 6) mitigates the impact of differing units or scales across criteria, ensuring consistency in evaluation. Through vector normalization, benefit-type criteria are positively scaled, while cost-type criteria are inversely normalized. This process enhances comparability across all criteria and prepares the data for the subsequent integration of criterion weights. Normalization represents a crucial transformation, establishing a dimensionless and standardized decision framework essential for robust multi-criteria analysis.

As seen in Table 7, the alternative software vendors are assessed through the evaluation stages. As a result of this analysis, ABC emerges as the top-performing software supplier, achieving the highest overall score.

Table 7. Scoring and ranking of alternatives.

Alternatives	Scores	Ranking
XYZ	0.4276	2
ABC	0.5724	1

4. Results, Discussion, and Implications

The software solutions adopted by businesses exert profound influence on strategic, managerial, and financial outcomes, with long-term implications for operational performance [18]. Given the substantial financial and operational investments required for software acquisition and implementation, stakeholders expect such decisions to yield sustainable competitive advantages. Within the transportation business, while diverse software solutions have emerged to address operational demands, the selection process remains inherently complex due to the multidimensional nature of evaluation criteria. Establishing a robust framework for identifying and prioritizing these criteria is therefore a critical prerequisite for informed decision making.

This study employs a systematic approach to compile and validate evaluation criteria through iterative consultations with business professionals (Tables 1 and 2). To enhance objectivity, the Best–Worst Method (BWM), a vector-based multi-criteria decision-making (MCDM) technique developed by [16], is selected for weighting criteria. The BWM reduces cognitive load by requiring only $2n - 3$ pairwise comparisons (versus $n(n - 1)/2$ in AHP), thereby improving reliability through minimized inconsistency [16]. To aggregate expert inputs, the Bayesian Best–Worst Method (BBWM) is applied, leveraging probability distributions to synthesize evaluations across multiple decision makers [64]. The resulting weights for main and sub-criteria are presented in Tables 3 and 4.

Functionality (weight: 0.2660, Rank 1) emerges as the paramount domain, underscoring its centrality in bridging software capabilities with operational demands. Such prioritization aligns with the literature positioning functionality as the cornerstone of TMS efficacy, particularly in dynamic logistics ecosystems where features such as load tracking (weight: 0.3584, Rank 1) and real-time data integration directly influence supply chain responsiveness [3,28]. The prominence of load tracking resonates with e-commerce-driven demands for shipment visibility, a critical driver of customer satisfaction (Sanders, 2016) [29]. Similarly, the weights assigned to software modules (0.3426, Rank 2) and customization (0.2990, Rank 3) reflect the business’s reliance on modular, adaptable solutions to address heterogeneous logistics workflows [14].

Cost (weight: 0.2270, Rank 2) follows closely, highlighting the business’s acute sensitivity to both initial expenditures and long-term financial viability. This finding mirrors studies emphasizing the strategic imperative to reconcile technological adoption with fiscal prudence in low-margin industries [1,7]. Within this domain, software license cost (0.3977, Rank 1) and update cost (0.3922, Rank 2) dominate, signaling concerns over legacy system maintenance and life cycle expenses [34]. The comparatively low weight of software module cost (0.2101, Rank 3) suggests a preference for holistic cost frameworks over fragmented pricing models. In this context, cloud-based solutions—characterized by lower upfront investments and scalable pricing models—may offer a viable alternative to traditional on-premises systems.

Technological competence (0.1952, Rank 3) and service (0.1939, Rank 4) exhibit near-parallel weights, reflecting their synergistic roles in ensuring system robustness and post-deployment support. Within the technological competence domain, information flow and transparency (0.3894, Rank 1) dominate, aligning with the literature stressing real-time data integration as a prerequisite for supply chain coordination [28,69]. In addition,

robust data integration, real-time information sharing, and system interoperability are essential for ensuring that a TMS seamlessly integrates with existing ERP or WMS systems, thereby streamlining workflows and reducing error rates. In addition, ease of use (0.3769, Rank 2) underscores the value of intuitive design in minimizing training overheads, a critical element in labor-intensive logistics operations [69]. Customized reporting (0.2337, Rank 3), though secondary, remains pertinent for businesses leveraging tailored analytics to monitor KPIs.

In the service domain, reliability (0.3560, Rank 1) emerges as the foremost priority, consistent with Parasuraman et al.'s (1988) [33] SERVQUAL framework, which prioritizes uninterrupted performance. After-sales support (0.2589, Rank 2) and training (0.2116, Rank 3) highlight the necessity of vendor assistance during transitional phases, while the marginal weight of error management (0.1734, Rank 4) suggests a preference for proactive system resilience over reactive troubleshooting.

Software developer (vendor) (0.1179, Rank 5) occupies the lowest priority amidst the domains, a seemingly counterintuitive result given the literature emphasizing vendor expertise [35]. However, such hierarchy likely reflects a pragmatic decision-making sequence: businesses prioritize immediate operational and financial criteria before evaluating vendor attributes as secondary differentiators [70]. Within this domain, industrial knowledge (0.5217, Rank 1) dominates, underscoring the value of domain-specific expertise in developing context-aware solutions [35]. References (0.2901, Rank 2) and reputation (0.1883, Rank 3) serve as validators of competency, though secondary to demonstrated expertise.

The application of TOPSIS to rank TMS software alternatives yields a clear differentiation in performance, as evidenced by the relative closeness coefficients presented in Table 7. ABC software (Rank 1) demonstrates superior alignment with the ideal solution compared to XYZ, indicating its enhanced capability to meet the prioritized evaluation criteria.

Overall, the methodology and findings provide logistics businesses with a robust framework for making informed TMS software decisions. In an environment where real-time tracking, data integration, cost control, and business-specific expertise are essential, selecting the appropriate TMS software is a strategic investment that directly impacts long-term operational success.

In addition to functional and financial considerations, this study underscores the transformative potential of TMS software in driving sustainable logistics. The integration of digital tools like TMSs supports environmentally responsible practices through real-time visibility, route efficiency, and intelligent demand forecasting—all of which contribute to lower emissions and resource optimization. In this context, digital transformation is not merely a technological shift but a pathway toward operational sustainability. Future research and practice should therefore prioritize software solutions that embed sustainability metrics, such as emissions dashboards or carbon cost calculators, into their core functionalities.

Based on these insights, the following practical recommendations can be made:

- **Operational Priorities:** Businesses should prioritize TMS solutions with robust load tracking and real-time data integration to meet evolving customer expectations.
- **Cost Strategy:** While upfront license costs are critical, long-term financial planning must account for updating expenses to avoid technological obsolescence.
- **Vendor Selection:** Vendor evaluations should balance industrial expertise with proven support capabilities, rather than relying solely on market reputation.

Finally, the methodological framework used in this study holds significant potential for extension across diverse logistical and operational contexts. Future research could expand the criteria hierarchy and apply alternative MCDM methodologies—such as Analytic Network Process (ANP), PROMETHEE, or ELECTRE—to evaluate software solutions within

specialized domains, including road, maritime, aviation, or rail transportation systems, warehousing operations, or logistics divisions of retail and manufacturing businesses.

Author Contributions: Validation, C.K.K.; Formal analysis, C.K.K., B.D.D. and G.A.; Investigation, C.K.K.; Resources, G.A.; Data curation, B.D.D.; Writing—original draft, B.D.D. and G.A. All authors have read and agreed to the published version of the manuscript.

Funding: We did not receive any partial or full funding to perform this study.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Christopher, M. *Logistics & Supply Chain Management*, 5th ed.; Pearson: London, UK, 2016.
2. Altuntas, G.; Yildirim, B.F. Logistics specialist selection with intuitionistic fuzzy TOPSIS method. *Int. J. Logist. Syst. Manag.* **2022**, *42*, 1–34.
3. Simchi-Levi, D.; Kaminsky, P.; Simchi-Levi, E. *Designing and Managing the Supply Chain: Concepts, Strategies, and Case Studies*, 5th ed.; McGraw-Hill Education: New York, NY, USA, 2021.
4. Kamble, S.S.; Gunasekaran, A.; Gawankar, S.A. Sustainable Business 4.0 framework: A systematic literature review identifying the current trends and future perspectives. *Process Saf. Environ. Prot.* **2018**, *117*, 408–425. [\[CrossRef\]](#)
5. Fawcett, S.E.; Ellram, L.M.; Ogden, J.A. *Supply Chain Management: From Vision to Implementation*; Pearson: London, UK, 2015.
6. Zsidisin, G.A.; Hartley, J.L.; Ogden, J.A. Supply risk perceptions and practices: An exploratory comparison of Brazilian and American business. *Int. J. Logist. Manag.* **2016**, *27*, 401–419.
7. Gunasekaran, A.; Papadopoulos, T.; Dubey, R.; Wamba, S.F.; Childe, S.J.; Hazen, B.; Akter, S. Big data and predictive analytics for supply chain and organizational performance. *J. Bus. Res.* **2017**, *70*, 308–317. [\[CrossRef\]](#)
8. Davenport, T.H. *The AI Advantage: How to Put the Artificial Intelligence Revolution to Work*; MIT Press: Cambridge, MA, USA, 2018.
9. Laudon, K.C.; Laudon, J.P. *Management Information Systems: Managing the Digital Business*, 17th ed.; Pearson: London, UK, 2020.
10. Somers, T.M.; Nelson, K. A taxonomy of players and activities across the ERP project life cycle. *Inf. Manag.* **2004**, *41*, 257–278. [\[CrossRef\]](#)
11. Wei, C.C.; Chien, C.F.; Wang, M.J.J. An AHP-based approach to ERP system selection. *Int. J. Prod. Econ.* **2005**, *96*, 47–62. [\[CrossRef\]](#)
12. Apak, S.; Gümüş, S.; Özkan, Ö. A new framework for ERP software selection decisions: An integrated approach. *J. Intell. Manuf.* **2016**, *27*, 549–562. [\[CrossRef\]](#)
13. Olayah, F.; Anaam, E.; Yahya, A.; Hamdi, M.; Shamsan, A.; Yahya, A.; Lateef, F.; Jawarneh, M. A systematic literature review for multiple-criteria decision-making approaches in E-CRM software. *Telematique* **2022**, *21*, 7444–7467.
14. Ghiani, G.; Laporte, G.; Musmanno, R. *Introduction to Logistics Systems Management*; Wiley: Hoboken, NJ, USA, 2013.
15. Crainic, T.G.; Gendreau, M.; Potvin, J.Y. Intelligent freight-transportation systems: Assessment and the contribution of operations research. *Transp. Res. Part C Emerg. Technol.* **2019**, *105*, 332–347. [\[CrossRef\]](#)
16. Rezaei, J. Best-worst multi-criteria decision-making method. *Omega* **2015**, *53*, 49–57. [\[CrossRef\]](#)
17. Mohammadi, M.; Rezaei, J. Bayesian best-worst method: A probabilistic group decision-making model. *Omega* **2020**, *96*, 102075. [\[CrossRef\]](#)
18. Calayoğlu, İ.; Yılmaz, R. Accounting for developed software in businesses for internal use on Turkish Accounting Standards. *J. Bus. Econ. Stud.* **2015**, *3*, 130–153.
19. Esteves, J.; Pastor, J. Enterprise resource planning systems research: An annotated bibliography. *Commun. AIS* **2001**, *7*, 1–52. [\[CrossRef\]](#)
20. Efe, B. A fuzzy multi-criteria decision framework for ERP system selection. *Int. J. Intell. Inf. Technol.* **2016**, *12*, 25–52. [\[CrossRef\]](#)
21. Konurhan, Z.; Yücesan, M.; Gül, M. Avalanche risk analysis by a combined geographic information system and Bayesian best-worst method. In *Advances in Best-Worst Method*; Rezaei, M., Ed.; Springer: Berlin/Heidelberg, Germany, 2023; pp. 193–210. [\[CrossRef\]](#)
22. Gulum Tas, P.; Sun, G. Emergency Service Quality Assessment Using SERVQUAL and BWM. In *Advances in Best-Worst Method*; Rezaei, M., Ed.; Springer: Berlin/Heidelberg, Germany, 2023; pp. 169–191. [\[CrossRef\]](#)

23. Amin-Tahmasbi, H.; Ayaran, A.; Zarepoor, M. Supplier selection for the oil business using a combined BWM & F-VIKOR, case study: National Iranian South Oil Business. In *Advances in Best-Worst Method*; Rezaei, M., Ed.; Springer: Berlin/Heidelberg, Germany, 2023; pp. 109–129. [\[CrossRef\]](#)
24. Mollaoglu, M.; Gul, M. A critical review of Best-Worst Method applications in the transportation and logistics business. *LogForum* **2025**, *21*, 77–95. [\[CrossRef\]](#)
25. Lieb, R.C.; Bentz, B.A. The use of third-party logistics services by large American manufacturers: The 2004 survey. *Transp. J.* **2005**, *44*, 5–15. [\[CrossRef\]](#)
26. Štemberger, M.I.; Škrinjar, R.; Hernaus, T. Organizational impacts of business process orientation. *TQM J.* **2009**, *21*, 387–402.
27. Carter, C.R.; Rogers, D.S.; Choi, T.Y. Toward the theory of the supply chain. *J. Supply Chain Manag.* **2015**, *51*, 89–97. [\[CrossRef\]](#)
28. Bowersox, D.J.; Closs, D.J.; Cooper, M.B. *Supply Chain Logistics Management*, 5th ed.; McGraw-Hill Education: New York, NY, USA, 2013.
29. Sanders, N.R. *Big Data Driven Supply Chain Management: A Framework for Implementing Analytics and Turning Information into Intelligence*; Pearson Education: London, UK, 2016.
30. Rushton, A.; Croucher, P.; Baker, P. *The Handbook of Logistics and Distribution Management*, 6th ed.; Kogan Page: London, UK, 2014.
31. Rai, A.; Patnayakuni, R.; Seth, N. Business performance impacts of digitally enabled supply chain integration capabilities. *MIS Q.* **2006**, *30*, 225–246. [\[CrossRef\]](#)
32. Zhu, K.; Kraemer, K.L.; Xu, S. The process of innovation assimilation by business in different countries: A technology diffusion perspective on e-business. *Manag. Sci.* **2006**, *52*, 1557–1576. [\[CrossRef\]](#)
33. Parasuraman, A.; Zeithaml, V.A.; Berry, L.L. SERVQUAL: A multiple-item scale for measuring consumer perceptions of service quality. *J. Retail.* **1988**, *64*, 12–40.
34. Kaplan, R.S.; Norton, D.P. *The Balanced Scorecard: Translating Strategy into Action*; Harvard Business Press: Brighton, MA, USA, 1996.
35. Lacity, M.C.; Willcocks, L.P. *Advanced Outsourcing Practice: Rethinking ITO, BPO, and Cloud Services*; Palgrave Macmillan: London, UK, 2012.
36. Smith, H.J.; Dinev, T.; Xu, H. Information privacy research: An interdisciplinary review. *MIS Q.* **2011**, *35*, 989–1015. [\[CrossRef\]](#)
37. Hidalgo, A.; Albors, J.; Hervás-Oliver, J.L. ICT and e-business: Their impact on innovation in manufacturing SMEs. *Int. J. Technol. Manag.* **2011**, *56*, 301–314.
38. Zaidan, A.A.; Zaidan, B.B.; Alaa, M.; Albahri, A.S. Evaluation and selection model for electronic health record software packages using multi-criteria decision-making techniques. *Comput. Biol. Med.* **2015**, *69*, 203–222.
39. Şen, A.; Baraçlı, H. A decision support model for ranking strategies based on resource allocation. *Gazi Univ. J. Sci.* **2010**, *23*, 395–405.
40. Omerali, H.; Kaya, T. The impact of enterprise resource planning (ERP) systems on business performance: Evidence from manufacturing companies. *J. Bus. Res.* **2021**, *124*, 345–356. [\[CrossRef\]](#)
41. Kannan, D.; Mina, H.; Nosrati-Abarghoee, S.; Khosrojerdi, G. Sustainable circular supplier selection: A novel hybrid approach. *Sci. Total Environ.* **2020**, *722*, 137936. [\[CrossRef\]](#) [\[PubMed\]](#)
42. Garg, C.P.; Shankar, R.; Agarwal, A. Analyzing the impact of digital transformation on sustainable supply chain performance using hybrid MCDM approach. *J. Clean. Prod.* **2022**, *330*, 129811. [\[CrossRef\]](#)
43. Özgül, O. An Enterprise Resource Planning Software Selection Process with Using TOPSIS and AHP Methods for a Business. Master's Thesis, Sakarya University, Serdivan, Türkiye, 2006.
44. Kaur, P.; Mahanti, N.C. A fuzzy ANP-based approach for selecting ERP systems. *Int. J. Soft Comput.* **2008**, *3*, 24–32.
45. Karsak, E.E.; Ozogul, C.O. An integrated decision making approach for ERP system selection. *Expert Syst. Appl.* **2009**, *36*, 660–667. [\[CrossRef\]](#)
46. Benlian, A.; Hess, T. Opportunities and risks of software-as-a-service: Findings from a survey of IT executives. *Decis. Support Syst.* **2011**, *52*, 232–246. [\[CrossRef\]](#)
47. Alpers, S.; Becker, C.; Eryilmaz, E.; Schuster, T. A systematic approach for evaluation and selection of ERP systems. In *Information Systems: Education, Applications, Research, Proceedings of the 7th SIGSAND/PLAIS EuroSymposium 2014, Gdansk, Poland, 25 September 2014*; Springer International Publishing: Berlin/Heidelberg, Germany, 2014; pp. 36–48.
48. Lai, K.-H.; Wong, C.W.Y.; Wang, X. Sustainability performance of logistics providers: A resource orchestration perspective. *Int. J. Prod. Econ.* **2022**, *247*, 108414. [\[CrossRef\]](#)
49. Zahedi, S.; Shahin, A.; Kazemi, A. Integration of fuzzy QFD and fuzzy MCDM for supplier ranking. *Int. J. Adv. Manuf. Technol.* **2011**, *56*, 689–702.
50. Tuş, A.; Adalı, E.A. A novel interval-valued intuitionistic fuzzy decision-making approach for digital transformation of smart cities. *Technol. Forecast. Soc. Change* **2019**, *147*, 119719. [\[CrossRef\]](#)

51. Zulkifli, N.; Abdullah, N.H.; Wahid, F.A. Evaluation of e-government implementation: A case study of Malaysia. In Proceedings of the International Conference on Information and Communication Technology for the Muslim World (ICT4M), Jakarta, Indonesia, 13–14 December 2010; IEEE: Piscataway, NY, USA, 2010; pp. 1–6. [\[CrossRef\]](#)
52. Çalışkan, C.; Kurşun, A.; Çalışkan, E. A hybrid MCDM methodology for ERP selection problem with interval type-2 fuzzy sets. *J. Intell. Fuzzy Syst.* **2019**, *37*, 6227–6238. [\[CrossRef\]](#)
53. Rouhani, B.D.; Ravasan, A.Z. A model for evaluating business intelligence success in organizations. *J. Intell. Stud. Bus.* **2014**, *4*, 37–54. Available online: <https://ojs.hh.se/index.php/JISIB/article/view/31> (accessed on 1 February 2025).
54. Şeker, S.; Kahraman, C. Evaluation of enterprise resource planning software packages using spherical fuzzy TOPSIS. *Expert Syst. Appl.* **2022**, *193*, 116516.
55. Khaled, B.; Idrissi, M.K. A framework for evaluating ERP implementation in SMEs in Morocco. In Proceedings of the International Conference on Information Technology and e-Services (ICITeS), Sousse, Tunisia, 24–26 March 2012; IEEE: Piscataway, NY, USA, 2011; pp. 1–6. [\[CrossRef\]](#)
56. Maram, A.; Abbas, A.; Abdalla, H. A hybrid MCDM approach for ERP system selection based on ANP and TOPSIS under fuzzy environment. *J. Intell. Fuzzy Syst.* **2019**, *36*, 4601–4615.
57. Okudan, G.E. A decision support system for selecting an enterprise resource planning (ERP) system for a small or medium-sized enterprise (SME). *Int. J. Prod. Res.* **2006**, *44*, 3691–3708.
58. Rouhani, S. Development of a business intelligence maturity model (BIMM) for strategic alignment. *J. Organ. End User Comput.* **2017**, *29*, 31–56. [\[CrossRef\]](#)
59. Kheybari, S.; Kazemi, M.; Rezaei, J. Bioethanol facility location selection using best-worst method. *Appl. Energy* **2019**, *242*, 612–623. [\[CrossRef\]](#)
60. Badri Ahmadi, H.; Kusi-Sarpong, S.; Rezaei, J. Assessing the social sustainability of supply chains using best best-worst method. *Resour. Conserv. Recycl.* **2017**, *126*, 99–106. [\[CrossRef\]](#)
61. Gupta, R.; Naqvi, S.K. A framework for applying CSFs to ERP software selection: An extension of fuzzy TOPSIS approach. *Int. J. Intell. Inf. Technol.* **2017**, *13*, 41–62. [\[CrossRef\]](#)
62. Rezaei, J.; Wang, J.; Tavasszy, L. Linking supplier development to supplier segmentation using Best Worst Method. *Expert Syst. Appl.* **2015**, *42*, 9152–9164. [\[CrossRef\]](#)
63. Altuntaş, G.; Semerciöz, F.; Sözüer, A. A model proposal for the identification and classification of criteria angel investors use to evaluate new initiatives. In Proceedings of the 27th National Management and Organization Congress, Antalya, Türkiye, 18–20 April 2019.
64. Ma, Y.; Wang, Y.; Wang, L.; Liu, Y. The impact of information technology on business performance: Evidence from China. *J. Syst. Inf. Technol.* **2019**, *21*, 42–58.
65. Hwang, C.L.; Yoon, K. *Multiple Attribute Decision Making: Methods and Applications*; Springer: Berlin/Heidelberg, Germany, 1981. [\[CrossRef\]](#)
66. Monjezi, M.; Shahriar, K.; Dehghani, H. Selection of optimum blasting pattern using multiple attribute decision making. *Int. J. Rock Mech. Min. Sci.* **2010**, *47*, 509–520. [\[CrossRef\]](#)
67. Shyji, K.; Ilangkumaran, M.; Kumanan, S. Multi-criteria decision-making approach to evaluate optimum maintenance strategy in textile business. *J. Qual. Maint. Eng.* **2008**, *14*, 375–386. [\[CrossRef\]](#)
68. Yurdakul, M.; İç, Y.T. Enterprise competitiveness using analytic network process: A case study. *J. Prod. Res.* **2003**, *7*, 11–29.
69. Wang, G.; Gunasekaran, A.; Ngai, E.W.T.; Papadopoulos, T. Big data analytics in logistics and supply chain management. *Int. J. Logist. Manag.* **2016**, *27*, 478–484. [\[CrossRef\]](#)
70. Doganer, D.B.; Altuntas, G. A Model Proposal For Identifying and Classifying Criteria Used in the Selection of Transportation Management System (TMS) Software. In Proceedings of the VIII. Transportation and Logistics National Congress, Zonguldak, Türkiye, 13–14 December 2024.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.

Reproduced with permission of copyright owner. Further reproduction
prohibited without permission.